

FERNADO GEORGE ANAK MANI

Kuching, Sarawak | +60 14-320 7322 | milobio351@gmail.com linkedin.com/in/fernado-george | github.com/Fernado03 | https://portfolio-2026-rho-lime.vercel.app/

PROFESSIONAL SUMMARY

Final-year Bachelor of Computer Science (Data Science) student (CGPA 3.75/4.00) with practical experience developing client-facing machine-learning, Generative AI, Retrieval-Augmented Generation (RAG), and computer-vision solutions. Built end-to-end Python systems spanning data preprocessing, retrieval, model inference, dashboards, and reporting. Dean's List recipient and Gold Medal winner in an AI innovation competition.

PROFESSIONAL EXPERIENCE

Breakfast Byte Sdn. Bhd. | AI/ML Development Intern

March 2026 - Present

Kuching, Sarawak

- Developed KI Konnekt for a German client, a Gradio-based AI assistant integrating local Ollama RAG and OpenAI workflows for document analysis, web search, structured outputs, file exports, image generation, and diagram generation.
- Engineered hybrid BM25/FAISS retrieval, multi-format document-processing pipelines, streaming responses, conversation persistence, usage tracking, and downloadable visual artifacts.
- Built an end-to-end Python computer-vision pipeline for a Sarawak car-wash client, converting daily CCTV footage into structured records for individual vehicle visits.
- Integrated vehicle-session detection, multi-bay assignment, license-plate OCR, vehicle-size and wash-action classification, temporal smoothing, rule-based package inference, and a local operator dashboard with traceable daily reports.

TECHNICAL SKILLS

- **Programming:** Python, SQL, R, Java, C++, JavaScript, HTML, CSS
- **Machine Learning and Data:** Data analysis, data preprocessing, feature engineering, machine learning, deep learning, model training and evaluation, computer vision, NLP, RAG, Large Language Models
- **Libraries and AI Platforms:** PyTorch, TensorFlow, OpenAI API, Gemini, Ollama, FAISS, BM25, Gradio
- **Web and Development:** React, Vite, Node.js, Tailwind CSS, Git, Docker, Google Cloud Platform, VS Code
- **Databases:** MongoDB, SQLite, relational SQL databases

EDUCATION

Universiti Malaysia Sabah | Bachelor of Computer Science (Data Science) with Honours **2022 - Present**

- Current CGPA: 3.75/4.00; Dean's List

Sarawak Matriculation College | Science Module II

2021 - 2022

- CGPA: 4.00/4.00; Best Computer Science Student Award

SELECTED PROJECTS AND ACHIEVEMENTS

Multimodal Emotion Recognition — Final Year Project | Best Research Award 2026

- Conducted a comparative study of contextual and statistical fusion strategies for robust multimodal emotion recognition using speech, visual, and text features.
- Developed and evaluated models using WavLM, DINOv2, ModernBERT, Bi-GRU, and ensemble-learning methods under Gaussian noise and missing-modality conditions.
- Achieved a 71.6% F1-score using an ensemble strategy, while the Contextual Bi-GRU architecture provided the strongest balance between robustness and computational efficiency.

KinaVis - AI Medical Scribe | Gold Medal - Festival Idea Sabah 2025

- Architected an AI-powered medical documentation system using Google Speech-to-Text, Gemini 2.5 Pro, and Retrieval-Augmented Generation.
- Implemented vector retrieval using all-MiniLM-L6-v2 to generate contextually grounded clinical records and reduce hallucinations.
- Reduced documentation time by approximately three to four hours per day and administrative costs by 30% during prototype evaluation.

MatchKami - AI School-NGO Matching Platform | 2nd Prize - JomHack Varsity Challenge 2025

- Developed an AI-powered matching system connecting under-resourced schools with suitable NGOs and donors.
- Implemented smart need-matching, a verified donor network, and an accessible AI chatbot on Telegram using Python and natural-language processing.
- Designed the solution to improve access to educational resources for underserved and rural communities.

LEADERSHIP AND CO-CURRICULAR INVOLVEMENT

PMFKI Student Association | Deputy Secretary (2022–2023) & Head of Academic and Career Affairs (2023–2024)

- Supported association administration, event coordination, communications, and academic and career-development activities for students.

Hackathon Team Lead and Lead Developer | 4+ Hackathons

- Led teams in more than four hackathons, including MarineHack and JomHack, while managing intensive 48-hour development sprints.
- Oversaw solution architecture, task delegation, development, team coordination, and final product pitching.
- Contributed to teams that secured one Gold Medal and one 2nd Prize.

REFERENCES

Dr. James Mountstephens

Senior Lecturer and Final-Year Project Supervisor
Faculty of Computing and Informatics, Universiti Malaysia Sabah
jmountstephens@ums.edu.my | +60 12-834 9044

Charissa Phua

Co-Founder, Director and Industrial Supervisor
Breakfast Byte Sdn. Bhd.
charissa.phua@breakfast-byte.com | +60 11-6744 1298